

Foundation Mathematics 1017SCG
Week 11 Workshop Solutions

Remember that the method shown in the solutions is not the only way to solve the question. There are usually multiple ways to solve the same question.

1. (a)

$$\begin{aligned}\int 4x^3 \, dx &= \frac{4x^4}{4} + C \\ &= x^4 + C\end{aligned}$$

(b)

$$\int x^5 \, dx = \frac{x^6}{6} + C$$

(c)

$$\begin{aligned}\int x - 2 \, dx &= \int x^1 - 2x^0 \, dx \\ &= \frac{x^2}{2} - 2x + C\end{aligned}$$

(d)

$$\begin{aligned}\int 3x - 8 \, dx &= \int 3x^1 - 8x^0 \, dx \\ &= \frac{3x^2}{2} - 8x + C\end{aligned}$$

(e)

$$\begin{aligned}\int 5x^2 - 6x + 7 \, dx &= \frac{5x^3}{3} - \frac{6x^2}{2} + 7x + C \\ &= \frac{5x^3}{3} - 3x^2 + 7x + C\end{aligned}$$

(f)

$$\begin{aligned}\int 9x^2 - 3x + 4 \, dx &= \frac{9x^3}{3} - \frac{3x^2}{2} + 4x + C \\ &= 3x^3 - \frac{3x^2}{2} + 4x + C\end{aligned}$$

(g)

$$\int 7x^3 - 4x^2 + 9x - 2 \, dx = \frac{7x^4}{4} - \frac{4x^3}{3} + \frac{9x^2}{2} - 2x + C$$

(h)

$$\int 5 \sin(x) + 6 \cos(x) \, dx = -5 \cos(x) + 6 \sin(x) + C$$

(i)

$$\int -4 \cos(x) + x \, dx = -4 \sin(x) + \frac{x^2}{2} + C$$

(j)

$$\int 2e^x \, dx = 2e^x + C$$

(k)

$$\int 6e^x + \sin(x) \, dx = 6e^x - \cos(x) + C$$

(l)

$$\int 7e^x + \frac{4}{x} \, dx = 7e^x + 4 \ln |x| + C$$

(m)

$$\int -\frac{6}{x} + 2 \, dx = -6 \ln |x| + 2x + C$$

2. When evaluating definite integrals, the solutions do **not** include the constant of integration. You are welcome to include the constant of integration in your working. It will give the same answer using either method.

(a)

$$\begin{aligned} \int_1^4 x \, dx &= \left[\frac{x^2}{2} \right]_1^4 \\ &= \left(\frac{4^2}{2} \right) - \left(\frac{1^2}{2} \right) \\ &= \frac{16}{2} - \frac{1}{2} \\ &= \frac{15}{2} \end{aligned}$$

(b)

$$\begin{aligned} \int_{-1}^3 x^3 \, dx &= \left[\frac{x^4}{4} \right]_{-1}^3 \\ &= \left(\frac{3^4}{4} \right) - \left(\frac{(-1)^4}{4} \right) \\ &= \frac{81}{4} - \frac{1}{4} \\ &= \frac{80}{4} \\ &= 20 \end{aligned}$$

(c)

$$\begin{aligned}\int_3^5 3x - 1 \, dx &= \left[\frac{3x^2}{2} - x \right]_3^5 \\&= \left(\frac{3 \times 5^2}{2} - 5 \right) - \left(\frac{3 \times 3^2}{2} - 3 \right) \\&= \frac{65}{2} - \frac{21}{2} \\&= 22\end{aligned}$$

(d)

$$\begin{aligned}\int_{-2}^3 2x^3 \, dx &= \left[\frac{2x^4}{4} \right]_{-2}^3 \\&= \left[\frac{x^4}{2} \right]_{-2}^3 \\&= \left(\frac{3^4}{2} \right) - \left(\frac{(-2)^4}{2} \right) \\&= \frac{81}{2} - \frac{16}{2} \\&= \frac{65}{2}\end{aligned}$$

(e)

$$\begin{aligned}\int_0^6 5 \, dx &= [5x]_0^6 \\&= (5 \times 6) - (5 \times 0) \\&= 30 - 0 \\&= 30\end{aligned}$$

(f)

$$\begin{aligned}\int_{-1}^1 x^2 + 3x - 1 \, dx &= \left[\frac{x^3}{3} + \frac{3x^2}{2} - x \right]_{-1}^1 \\&= \left(\frac{1^3}{3} + \frac{3 \times 1^2}{2} - 1 \right) - \left(\frac{(-1)^3}{3} + \frac{3 \times (-1)^2}{2} - (-1) \right) \\&= \left(\frac{1}{3} + \frac{3}{2} - 1 \right) - \left(-\frac{1}{3} + \frac{3}{2} + 1 \right) \\&= \frac{5}{6} - \frac{13}{6} \\&= -\frac{8}{6} \\&= -\frac{4}{3}\end{aligned}$$

(g)

$$\begin{aligned}\int_{-2}^3 2 + 3x^2 \, dx &= \left[2x + \frac{3x^3}{3} \right]_{-2}^3 \\&= [2x + x^3]_{-2}^3 \\&= (2 \times 3 + 3^3) - (2 \times -2 + (-2)^3) \\&= 33 - -12 \\&= 45\end{aligned}$$

(h)

$$\begin{aligned}\int_0^{\frac{\pi}{2}} 3 \cos(x) \, dx &= [3 \sin(x)]_0^{\frac{\pi}{2}} \\&= \left(3 \sin\left(\frac{\pi}{2}\right) \right) - (3 \sin(0)) \\&= (3 \times 1) - (3 \times 0) \\&= 3 - 0 \\&= 3\end{aligned}$$

(i)

$$\begin{aligned}\int_{-2}^3 4e^x + 2x \, dx &= \left[4e^x + \frac{2x^2}{2} \right]_{-2}^3 \\&= [4e^x + x^2]_{-2}^3 \\&= (4e^3 + 3^2) - (4e^{-2} + (-2)^2) \\&= 4e^3 + 9 - 4e^{-2} - 4 \\&= 84.80\end{aligned}$$

(j)

$$\begin{aligned}\int_3^5 \frac{6}{x} \, dx &= [6 \ln |x|]_3^5 \\&= 6 \ln(5) - 6 \ln(3) \\&= 3.06\end{aligned}$$

(k)

$$\begin{aligned}\int_0^2 2e^x - x^2 \, dx &= \left[2e^x - \frac{x^3}{3} \right]_0^2 \\&= \left(2e^2 - \frac{2^3}{3} \right) - \left(2e^0 - \frac{0^3}{3} \right) \\&= \left(2e^2 - \frac{8}{3} \right) - (2 - 0) \\&= 2e^2 - \frac{8}{3} - 2 \\&= 10.11\end{aligned}$$

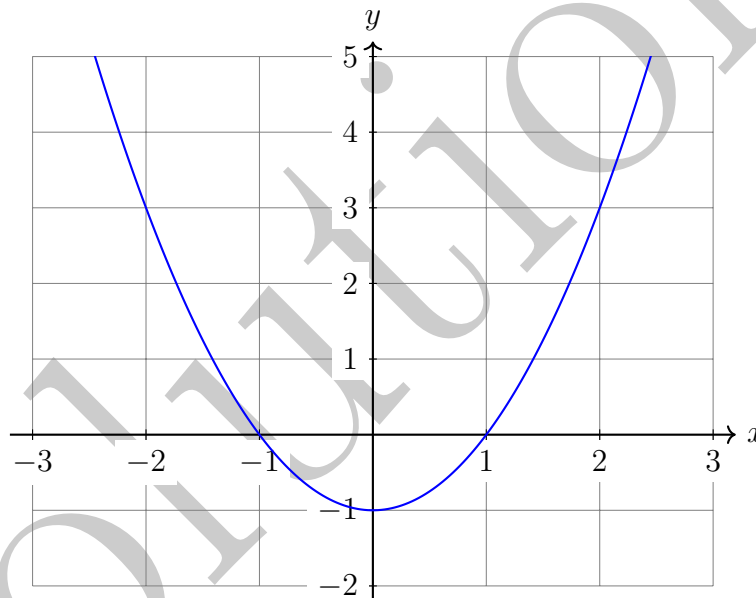
(1)

$$\begin{aligned}\int_{-\pi}^{\frac{3\pi}{2}} -\sin(x) \, dx &= [\cos(x)]_{-\pi}^{\frac{3\pi}{2}} \\ &= \cos\left(\frac{3\pi}{2}\right) - \cos(-\pi) \\ &= 0 - (-1) \\ &= 1\end{aligned}$$

3. (a) The graph of $y = x^2 - 1$ is shown below. See Chapter 7 for additional information on how to graph a quadratic function.

- The x -intercepts are $x = -1$ and $x = 1$
- The y -intercept is $y = -1$
- The turning point is at $x = 0, y = -1$

As mentioned in the question, it is important that your graph is accurate for $0 \leq x \leq 2$ as this will be needed for later parts in the question.



(b) When evaluating a definite integral, the area above the x -axis is considered ‘positive area’ whereas the area below the x -axis is considered ‘negative area’. Therefore, when using a definite integral, area above the x -axis ‘cancels’ with area below the x -axis.

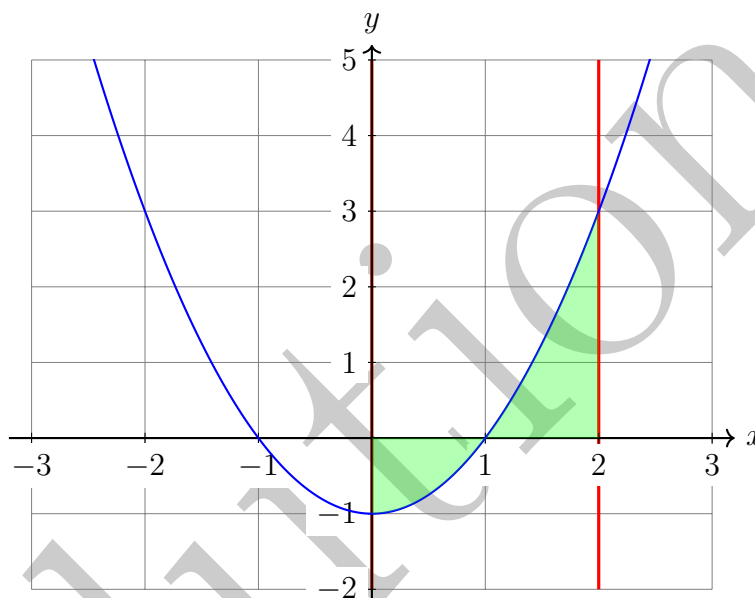
To find the total area bounded by a curve, the area above the x -axis must be calculated separately to the area below the x -axis. The two areas can then be combined to give the total area bounded by the curve.

For the function $y = x^2 - 1$, the function is negative for $0 \leq x < 1$ and positive for $1 < x \leq 2$. Therefore the definite integral from $x = 0$ to $x = 2$ will not give the total area bounded by the curve between $x = 0$ and $x = 2$.

(c)

$$\begin{aligned}\int_0^2 x^2 - 1 \, dx &= \left[\frac{x^3}{3} - x \right]_0^2 \\&= \left(\frac{2^3}{3} - 2 \right) - \left(\frac{0^3}{3} - 0 \right) \\&= \frac{2}{3} - 0 \\&= \frac{2}{3}\end{aligned}$$

(d) The area bounded by the curve $y = x^2 - 1$ and the x -axis between $x = 0$ and $x = 2$ is shaded in green on the graph below.



We will now find the area of the shaded region below the x -axis.

$$\begin{aligned}\int_0^1 x^2 - 1 \, dx &= \left[\frac{x^3}{3} - x \right]_0^1 \\&= \left(\frac{1^3}{3} - 1 \right) - \left(\frac{0^3}{3} - 0 \right) \\&= -\frac{2}{3}\end{aligned}$$

Therefore, the area of the shaded region below the x -axis is $\frac{2}{3}$ units².

We will now find the area of the shaded region above the x -axis.

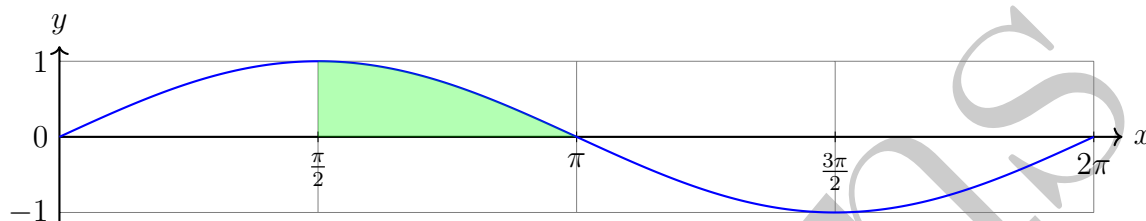
$$\begin{aligned}\int_1^2 x^2 - 1 \, dx &= \left[\frac{x^3}{3} - x \right]_1^2 \\&= \left(\frac{2^3}{3} - 2 \right) - \left(\frac{1^3}{3} - 1 \right) \\&= \frac{2}{3} + \frac{2}{3} \\&= \frac{4}{3}\end{aligned}$$

Therefore the area of the shaded region below the x -axis is $\frac{4}{3}$ units². The total area of the shaded region is the sum of the two areas found above.

$$\frac{2}{3} + \frac{4}{3} = \frac{6}{3} = 2$$

Therefore the area bounded by the curve $y = x^2 - 1$ and the x -axis between $x = 0$ and $x = 2$ is 2 units².

4. (a) A graph of $y = \sin(x)$ is shown below. The area that needs to be calculated is shaded green.

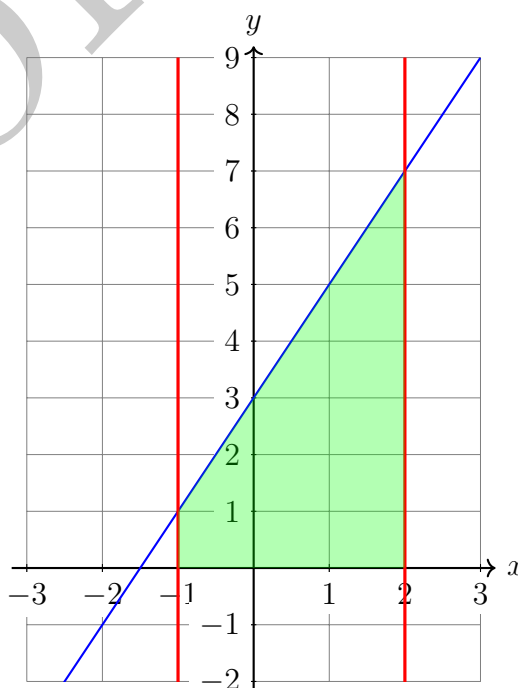


As $y = \sin(x)$ is positive (or zero) between $x = \frac{\pi}{2}$ and $x = \pi$, the shaded area can be calculated using a single definite integral.

$$\begin{aligned} \int_{\frac{\pi}{2}}^{\pi} \sin(x) \, dx &= [-\cos(x)]_{\frac{\pi}{2}}^{\pi} \\ &= (-\cos(\pi)) - \left(-\cos\left(\frac{\pi}{2}\right)\right) \\ &= (1) - (0) \\ &= 1 \end{aligned}$$

Therefore the area bounded by $y = \sin(x)$ and the x -axis between $x = \frac{\pi}{2}$ and $x = \pi$ is 1 unit².

- (b) A graph of $y = 2x + 3$ is shown below. The area that needs to be calculated is shaded green.

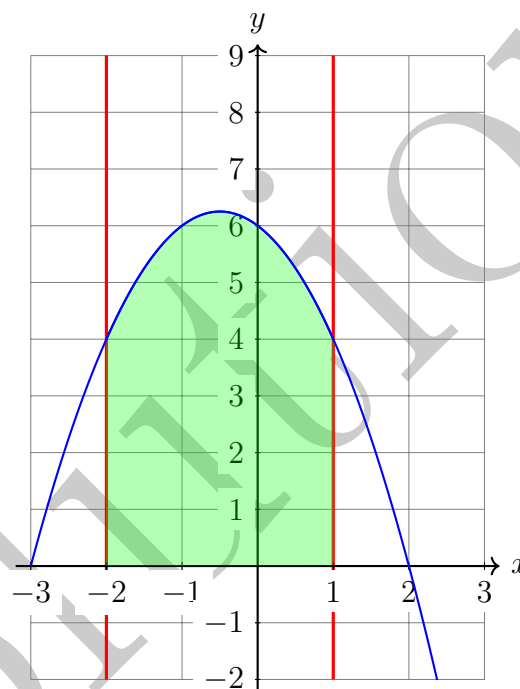


As $y = 2x + 3$ is positive between $x = -1$ and $x = 2$, the shaded area can be calculated using a single definite integral.

$$\begin{aligned}\int_{-1}^2 2x + 3 \, dx &= \left[\frac{2x^2}{2} + 3x \right]_{-1}^2 \\ &= [x^2 + 3x]_{-1}^2 \\ &= (2^2 + 3 \times 2) - ((-1)^2 + 3 \times (-1)) \\ &= (10) - (-2) \\ &= 12\end{aligned}$$

Therefore the area bounded by $y = 2x + 3$ and the x -axis between $x = -1$ and $x = 2$ is 12 units².

- (c) A graph of $y = -x^2 - x + 6$ is shown below. The area that needs to be calculated is shaded green.

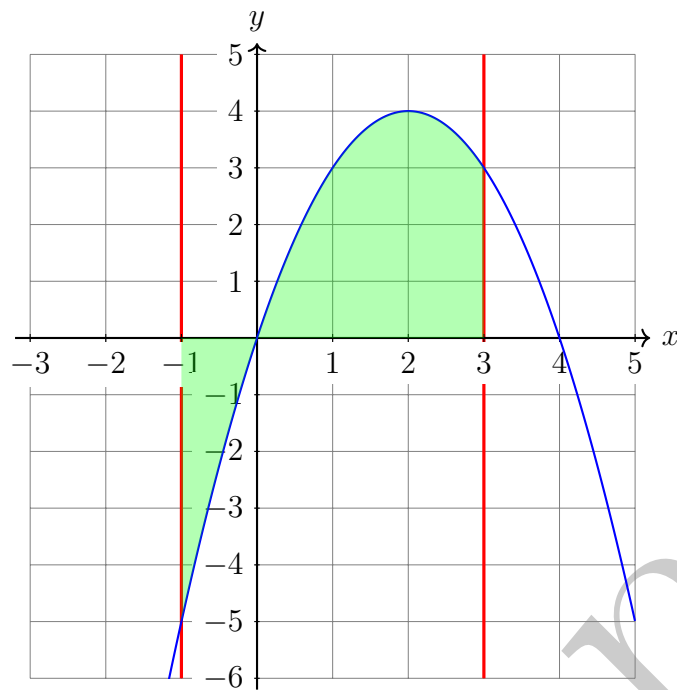


As $y = -x^2 - x + 6$ is positive between $x = -2$ and $x = 1$, the shaded area can be calculated using a single definite integral.

$$\begin{aligned}\int_{-2}^1 6 - x - x^2 \, dx &= \left[6x - \frac{x^2}{2} - \frac{x^3}{3} \right]_{-2}^1 \\ &= \left(6 \times 1 - \frac{1^2}{2} - \frac{1^3}{3} \right) - \left(6 \times -2 - \frac{(-2)^2}{2} - \frac{(-2)^3}{3} \right) \\ &= \frac{31}{6} - -\frac{34}{3} \\ &= \frac{33}{2}\end{aligned}$$

Therefore the area bounded by $y = 6 - x - x^2$ and the x -axis between $x = -2$ and $x = 1$ is $\frac{33}{2}$ units².

- (d) A graph of $y = -x^2 + 4x$ is shown below. The area that needs to be calculated is shaded green. It is important to note that some of the shaded area is below the x -axis.



We will now find the area of the shaded region that is below the x -axis.

$$\begin{aligned}
 \int_{-1}^0 -x^2 + 4x \, dx &= \left[-\frac{x^3}{3} + \frac{4x^2}{2} \right]_{-1}^0 \\
 &= \left[-\frac{x^3}{3} + 2x^2 \right]_{-1}^0 \\
 &= \left(\frac{0^3}{3} + 2 \times 0^2 \right) - \left(-\frac{(-1)^3}{3} + 2 \times (-1)^2 \right) \\
 &= (0) - \left(\frac{7}{3} \right) \\
 &= -\frac{7}{3}
 \end{aligned}$$

Therefore, the area of the shaded region below the x -axis is $\frac{7}{3}$ units². We will now find the area of the shaded region that is above the x -axis.

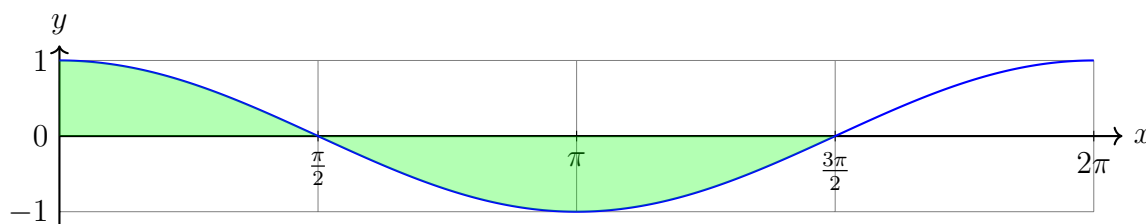
$$\begin{aligned}
 \int_0^3 -x^2 + 4x \, dx &= \left[-\frac{x^3}{3} + \frac{4x^2}{2} \right]_0^3 \\
 &= \left[-\frac{x^3}{3} + 2x^2 \right]_0^3 \\
 &= \left(-\frac{3^3}{3} + 2 \times 3^2 \right) - \left(-\frac{0^3}{3} + 2 \times 0^2 \right) \\
 &= 9 - 0 \\
 &= 9
 \end{aligned}$$

Therefore, the area of the shaded region above the x -axis is 9 units². The total area of the shaded region is the sum of the two areas found above.

$$\frac{7}{3} + 9 = \frac{34}{3}$$

Therefore the area bounded by $y = -x^2 + 4x$ and the x -axis between $x = -1$ and $x = 3$ is $\frac{34}{3}$ units².

- (e) A graph of $y = \cos(x)$ is shown below. The area that needs to be calculated is shaded green. It is important to note that some of the shaded area is below the x -axis.



We will now find the area of the shaded region that is above the x -axis.

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \cos(x) \, dx &= [\sin(x)]_0^{\frac{\pi}{2}} \\ &= \sin\left(\frac{\pi}{2}\right) - \sin(0) \\ &= 1 - 0 \\ &= 1\end{aligned}$$

Therefore, the area of the shaded region above the x -axis is 1 unit². We will now find the area of the shaded region that is below the x -axis.

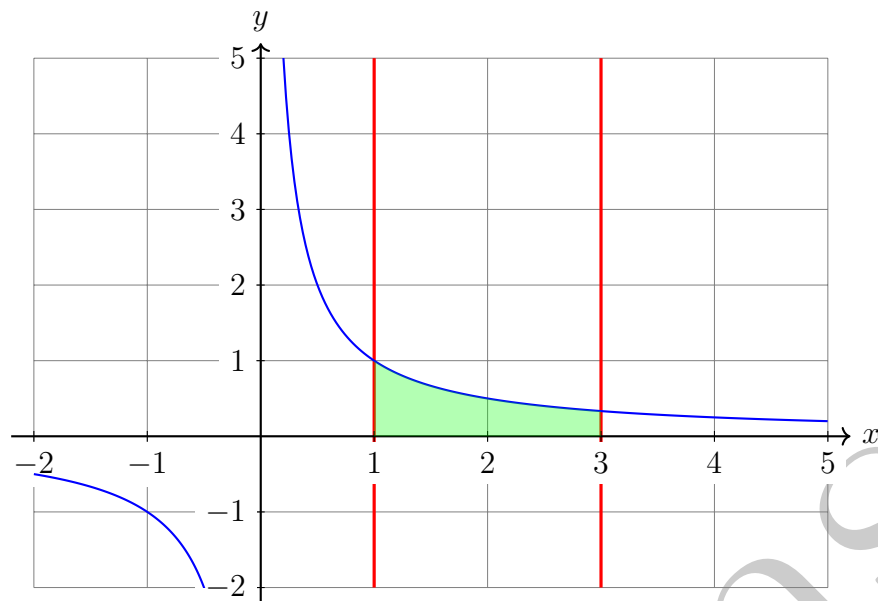
$$\begin{aligned}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \cos(x) \, dx &= [\sin(x)]_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \\ &= \sin\left(\frac{3\pi}{2}\right) - \sin\left(\frac{\pi}{2}\right) \\ &= -1 - 1 \\ &= -2\end{aligned}$$

Therefore the area of the shaded region below the x -axis is 2 units². The total area of the shaded region is the sum of the two areas found above.

$$1 + 2 = 3$$

Therefore the area bounded by $y = \cos(x)$ and the x -axis between $x = 0$ and $x = \frac{3\pi}{2}$ is 3 units².

- (f) A graph of $y = \frac{1}{x}$ is shown below. The area that needs to be calculated is shaded green.



As $y = \frac{1}{x}$ is positive between $x = 1$ and $x = 3$, the area under the curve can be calculated using a single definite integral.

$$\begin{aligned}\int_1^3 \frac{1}{x} dx &= [\ln |x|]_1^3 \\ &= \ln(3) - \ln(1) \\ &= \ln(3) - 0 \\ &= \ln(3)\end{aligned}$$

Therefore, the area bounded by $y = \frac{1}{x}$ and the x -axis between $x = 1$ and $x = 3$ is approximately 1.10 units².

5. (a)

$$\begin{aligned}\int_0^1 4t^3 + 12t^2 + 75 dt &= \left[\frac{4t^4}{4} + \frac{12t^3}{3} + 75t \right]_0^1 \\ &= [t^4 + 4t^3 + 75t]_0^1 \\ &= (1^4 + 4 \times 1^3 + 75 \times 1) - (0^4 + 4 \times 0^3 + 75 \times 0) \\ &= 80 - 0 \\ &= 80\end{aligned}$$

Therefore during the first minute of exercise, the heart beats 80 times.

- (b) During the first minute of exercise, the heart beats 80 times with each beat pumping 72mL of blood.

$$80 \times 72 = 5760\text{mL}$$

During the first minute of exercise, the heart pumps 5760mL of blood.

- (c) We will start by calculating the number of heart beats during the second minute of

exercise.

$$\begin{aligned}\int_1^2 4t^3 + 12t^2 + 75 \, dt &= \left[\frac{4t^4}{4} + \frac{12t^3}{3} + 75t \right]_1^2 \\&= [t^4 + 4t^3 + 75t]_1^2 \\&= (2^4 + 4 \times 2^3 + 75 \times 2) - (1^4 + 4 \times 1^3 + 75 \times 1) \\&= 198 - 80 \\&= 118\end{aligned}$$

During the second minute of exercise, the heart beats 118 times with each beat pumping 72mL of blood.

$$118 \times 72 = 8496\text{mL}$$

During the second minute of exercise, the heart pumps 8496mL of blood.

(d)

$$5760 + 8496 = 14256$$

Therefore, during the first two minutes of exercise, the heart pumps 14256mL of blood.